

Factor Risk Model — Summary

A short summary ahead of Monday. The full detail is in the attached document (`factor-model-roadmap.html`) — this page is the gist.

This is a proof-of-concept Barra-style equity factor-risk model. It runs on free, public data and is demoed against the Soros 13F book. I've built it from scratch, ahead of joining — building each piece myself is how I'm learning the model, and how I'll be useful sooner once I start.

Chris's input has been pivotal. He's course-corrected the parts where I don't yet have the depth, and much of what's here came from his steers. The quality is in the details: how loadings are built, how the estimation universe is chosen, how descriptors are standardised, and how missing data is handled. That's where the work has gone.

How it's put together

Two halves that talk only through a small set of data files:

- **Builders** pull the raw inputs (SEC 13F, EDGAR fundamentals, public prices) and write the data frames — loadings, factor returns, specific risk, positions.
- An **Atoti cube** reads those and exposes the exposures plus one scenario / stress engine — historical simulation, event replay, hypothetical shocks.
- On top sits a **risk-tooling layer** (limits, data-quality checks, VaR backtest, trends, stress, pre-trade what-if, liquidity, universe diagnostics) and a small layer that reads a view's numbers back in plain English.

A good part of that risk-tooling layer is already live — VaR, ES, stress, backtesting, liquidity. One caveat: these are the standard measures. They're where I learned the mechanics, but I'd guess they're not central to how you manage risk at Soros. The review with Chris has been about moving past them — getting the factor model itself right, and validating it against realised PnL.

Two known limits worth naming. Country is hardcoded to US — the dimension is wired, it just needs a domicile / listing source to light up. And the estimation universe is capped at 250 names so the demo runs quickly; factor-return quality scales with how many names you fit across, so widening the cap improves the fit. It's a sizing choice, not a limit of the approach.

Where it stands — the review items

Each point Chris raised, with status:

#	ITEM	STATUS	NOTE
1	Point-in-time data	● IN PLACE	survivorship-free; bias-free estimation
2-3	Estimation vs. coverage universe; cap estimation, not coverage	● DONE – SHIPPED	the keystone — off-index names now read their true loadings; clean factor fit
4	Drawdown lens	● DONE – SHIPPED	the path risk VaR/ES miss
5	Active risk & return vs. a benchmark	● AGREED – NEXT	benchmark agreed: MSCI Large Cap; unblocks tracking error & active return
6	Intentional vs. unwanted loadings (hedging)	● EVIDENCE SHIPPED	style-drift attribution — rotation vs. re-pricing; final call needs desk intent
7	Orthogonalisation & multicollinearity	● PARTIAL	builder change
8	Linked securities (share classes / ADRs)	● PROPOSED	block-diagonal specific risk
9	PnL attribution & factor-model validation	● DESIGNED – ENDORSED	realised PnL by factor + residual; residual-correlation tests; linked to the ex-ante risk decomposition

- In place ● Done — built for this review ● Designed / partial — awaiting sign-off
- Proposed

What's next

The foundations are mostly in. The open points:

- **PnL attribution & factor-model validation (§9)** — the natural next build, and more relevant than the standalone measures. It splits realised PnL into a factor part and a residual, checks whether the residual is large and correlated, and reads each result against the risk expected at the start of the period. Designed and endorsed by Chris.
- **Active risk vs. a benchmark (§5)** — benchmark agreed (MSCI Large Cap), so this is ready to build: tracking error and active return.
- **What's most useful to the desk** — I'd value your view on which of these answers a real question, so I build for that.

IN ONE LINE

Building this from scratch is how I'm learning the model ahead of joining, and Chris's steers have been pivotal. The standard risk measures are live but probably not central to how Soros runs risk — the real work is getting the factor model right and validating it against realised PnL. The benchmark for active risk is agreed: MSCI Large Cap.

Full detail and the methodology are in the attached `factor-model-roadmap.html`.